RETINAL DISEA

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Retinal diseases = HIGH yield in theory + MCQs!
Tables are your weapon!

Retina | DR | ARMD | Ophthalmology

WHAT'S INSIDE

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DR = #1 cause of blindness in working age group! Super HY!

ETDRS Classification of Diabetic Retinopathy

Stage	Features
Mild NPDR	Microaneurysms only
Moderate NPDR	Microaneurysms + dot/blot hemorrhages + hard exudates + CWS
Severe NPDR	4-2-1 rule: 4 quadrants hemorrhages, 2 venous beading, 1 IRMA
Very severe NPDR	Any 2 of the above 4-2-1 criteria
Early PDR	New vessels (NVD/NVE) without high-risk features
High-risk PDR	NVD >1/3 DD OR NVD + VH OR NVE + VH

★ **4-2-1 Rule for Severe NPDR**
4 quadrants hemorrhages + 2 quadrants venous beading + 1 quadrant IRMA
50% severe NPDR -> PDR in 1 year; 75% very severe NPDR -> PDR in 1 year

NPDR vs PDR — Key Differences

Feature	NPDR	PDR
Neovascularization	Absent	Present (hallmark)
Hemorrhages	Dot, blot, flame	Vitreous/preretinal
Main risk	Macular edema	Vitreous hemorrhage, TRD
Treatment	Control DM, anti-VEGF for DME	PRP laser + anti-VEGF

- DME (center-involving): Anti-VEGF (Bevacizumab, Ranibizumab, Aflibercept)
- DME (non-center): Focal/grid laser (ETDRS guidelines)
- PDR: Panretinal photocoagulation (PRP) + anti-VEGF; Vitrectomy if VH/TRD

ARMD = #1 cause of blindness in elderly in developed countries!

Age-Related Macular Degeneration (ARMD)

Feature	Dry (Non-exudative)	Wet (Exudative)
Frequency	90% of cases	10% but causes 90% blindness
Hallmark	Drusen + RPE atrophy	CNVM (choroidal neovascular membrane)
Progression	Slow, gradual	Rapid visual loss
Treatment	AREDS vitamins (C,E,Zn,Cu)	Anti-VEGF injections (1st line)
Key sign	Geographic atrophy	Subretinal hemorrhage/fluid

★ **Drusen = yellow deposits beneath RPE. Hard drusen (small) = benign; Soft drusen (large) = risk for wet ARMD.**

Retinal Detachment — Types

Type	Mechanism	Causes
Rhegmatogenous	Retinal break/tear	Myopia, PVD, trauma (most common)
Tractional	Vitreoretinal traction	PDR, ROP, PVR
Exudative	Subretinal fluid	Tumor, inflammation, VKH, toxemia

- Flashes of light (photopsia) — vitreoretinal traction
- Floaters — vitreous hemorrhage or pigment cells (Shafer's sign)
- Curtain/shadow falling over vision — progressive field loss
- Painless sudden visual loss — macula off detachment

Retinal Detachment — Management

Procedure	Indication
Laser barrage / cryopexy	Retinal tears/holes without detachment
Scleral buckling	Rhegmatogenous RD (conventional surgery)
Pars plana vitrectomy	Complex RD, tractional, giant tears, VH
Pneumatic retinopexy	Superior, simple rhegmatogenous RD

CRAO vs CRVO = classic comparison question! Know the fundus signs!

CRAO vs CRVO — The Comparison

Feature	CRAO	CRVO
Type	Arterial occlusion	Venous occlusion
Onset	Sudden painless loss	Sudden painless loss
Fundus	Pale retina + cherry red spot	Blood & thunder fundus
Hemorrhages	Few/absent	Flame-shaped in all 4 quadrants
Cotton wool spots	Few	Many (ischemic type)
Edema	Retinal edema (pale)	Disc edema, macular edema
RAPD	Present	May be present (ischemic)
Risk factors	Emboli, GCA, carotid disease	HTN, DM, glaucoma, hyperviscosity
Emergency?	YES — 90 min golden period	No (but urgent referral)

★ **CRAO = OPHTHALMIC EMERGENCY! Central retinal artery = end artery.**
Cherry red spot = fovea (supplied by choroid) looks red against pale ischemic retina.
Rx: ocular massage, AC paracentesis, IV acetazolamide, vasodilators

CRVO — Non-ischemic vs Ischemic

Feature	Non-ischemic	Ischemic
Vision	6/12 to 6/60	< 6/60, often CF
Hemorrhages	Moderate	Severe, all quadrants
RAPD	Absent	Present
Complication	Macular edema	Neovascular glaucoma (100-day glaucoma)
Treatment	Anti-VEGF for ME	PRP for NV + anti-VEGF

★ **100-Day Glaucoma**
Ischemic CRVO -> neovascular glaucoma develops ~100 days later. Monitor for NVI (rubeosis iridis)!

Retinoblastoma = most common intraocular tumor of childhood. Must know!

Retinoblastoma — Key Facts

Feature	Detail
Definition	Most common primary intraocular malignancy of childhood
Gene	RB1 gene on chromosome 13q14 (tumor suppressor)
Inheritance	40% hereditary (bilateral), 60% sporadic (unilateral)
Presentation	Leukocoria (white pupillary reflex) — most common sign
Other signs	Strabismus, secondary glaucoma, pseudohypopyon
Diagnosis	B-scan USG (calcification), CT/MRI orbit, fundoscopy
Trilateral Rb	Bilateral Rb + pineoblastoma (intracranial)

Classification + Management

Stage / Group	Management
Group A-B (small)	Focal therapy — laser, cryotherapy, thermotherapy
Group C-D (advanced)	Chemoreduction + focal therapy
Group E (end-stage)	Enucleation (eye removal)
Extraocular	Chemotherapy + radiation + enucleation
Metastatic	High-dose chemo + stem cell rescue

Differential Diagnosis of Leukocoria

Condition	Key Feature
Retinoblastoma	Most important — calcification on CT
Persistent fetal vasculature (PFV)	Unilateral, microphthalmos, no calcification
Coat's disease	Unilateral, exudative RD, boys
ROP (Retinopathy of prematurity)	Bilateral, premature, low birth weight
Congenital cataract	Lens opacity, no retinal mass
Toxocara endophthalmitis	Unilateral, granuloma, ELISA positive

Retina = 15-20 marks in every exam! You've GOT this, Sammy!

Most Repeated LAQs (10-12 marks)

- 1. Classify diabetic retinopathy (ETDRS). Describe management of PDR
- 2. Compare CRAO and CRVO. Describe management of CRAO
- 3. Classify retinal detachment. Describe management of RD
- 4. Describe ARMD — types, features, management
- 5. Write about retinoblastoma — presentation, diagnosis, management

Most Repeated SAQs (5-6 marks)

- 1. NPDR vs PDR comparison (table)
- 2. Cherry red spot — causes and significance
- 3. Leukocoria — differential diagnosis
- 4. 100-day glaucoma — explain
- 5. Anti-VEGF agents in ophthalmology
- 6. Panretinal photocoagulation — indications

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Cherry red spot at macula	CRAO (also Tay-Sachs, Niemann-Pick)
Blood and thunder fundus	CRVO
4-2-1 rule	Severe NPDR (ETDRS)
100-day glaucoma	Ischemic CRVO -> NVG
Most common cause of leukocoria	Retinoblastoma
RB1 gene — chromosome	13q14 (tumor suppressor)
Sea-fan neovascularization	Sickle cell retinopathy
Drusen	ARMD (dry type)
Anti-VEGF first-line for DME	Ranibizumab / Bevacizumab / Aflibercept
Shafer's sign (tobacco dust)	Rhegmatogenous retinal detachment
Bull's eye maculopathy	Chloroquine toxicity

♥ **The retina is where vision begins, Sammy! Every detail you learn here** helps you save someone's sight. Keep shining, future doctor!